

KAHULUI HARBOR, HI, USA

WMO: 997384

Lat: 20.8950N

Lon: 156.4690W

Elev: 4

StdP: 101.28

Time Zone: -10.00 (W10)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
3	17.7	18.7	N/A	N/A	N/A	N/A	N/A	N/A	12.6	23.2	11.7	23.3	1.5	160	0.432

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
9	3.1	28.5	N/A	27.9	N/A	27.5	N/A	N/A	N/A	N/A	N/A	N/A	N/A	6.2	50

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
11.6	10.7	9.5	DB	15.7	29.4	0.5	1.0	15.3	30.1	15.0	30.7	14.7	31.2	14.3	31.9	
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	23.7	22.1	22.1	22.1	22.8	23.6	24.3	24.9	25.3	25.3	25.0	24.0	23.0
	DBStd	1.58	1.12	1.21	1.09	1.01	0.91	0.69	0.90	0.89	1.08	0.96	1.09	1.10
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD10.0	5014	376	340	374	385	423	429	463	474	460	465	421	403
	CDD18.3	1972	118	107	116	135	164	179	204	216	210	207	171	145
	CDH23.3	9808	253	224	146	321	627	878	1324	1552	1624	1514	877	468
	CDH26.7	353	1	1	1	1	7	13	38	75	117	71	23	4
(14) Wind	WSAvg	5.2	4.3	4.5	5.3	5.5	5.3	6.1	6.0	5.7	5.3	4.8	4.9	4.4
(15) Precipitation	PrecAvg	700	150	100	96	53	31	10	12	16	13	46	88	85
	PrecMax	1136	568	340	330	196	122	72	55	40	44	208	357	314
	PrecMin	275	18	6	2	1	0	0	0	1	1	0	3	5
	PrecStd	270	140	95	80	51	36	14	12	11	12	53	95	78
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	26.8	26.8	26.5	26.7	27.5	28.0	28.6	29.0	29.5	28.9	28.3	27.3
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	DB	25.9	26.0	25.3	25.9	26.7	26.9	27.7	28.0	28.5	28.0	27.4	26.5
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	DB	25.3	25.3	24.7	25.4	26.1	26.3	27.1	27.5	27.9	27.5	26.7	25.8
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	DB	24.7	24.7	24.1	24.9	25.6	25.8	26.7	27.0	27.4	26.9	26.2	25.3
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(35) Mean Daily Temperature Range	5% DB	MDBR	4.8	4.2	3.4	3.5	3.7	2.9	2.8	2.8	3.1	3.7	3.8	4.4
		MCDBR	6.0	5.6	4.5	4.5	4.4	3.7	3.3	3.5	3.8	4.2	4.7	5.4
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(40) Clear-Sky Solar Irradiance	taub		0.387	0.395	0.407	0.420	0.410	0.395	0.396	0.396	0.386	0.397	0.397	0.379
	taud		2.485	2.460	2.434	2.377	2.440	2.482	2.489	2.489	2.516	2.499	2.488	2.532
	Ebn at Noon		886	900	900	887	886	891	891	896	906	886	868	882
	Edh at Noon		101	110	117	125	116	111	110	110	106	104	100	94
(44) All-Sky Solar Radiation	RadAvg		4.49	5.05	5.66	6.46	6.88	7.08	7.00	6.80	6.37	5.39	4.55	4.10
	RadStd		0.19	0.30	0.53	0.38	0.35	0.18	0.17	0.28	0.18	0.14	0.11	0.35

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (11 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+90	+90

Nomenclature: See separate page